APPROVALS MAY BE VIEWED IN PLM SYSTEM					
REV	ECO	DESCRIPTION	DRAWN BY	APPROWED BY	DATE

ELECTRONIC APPROVALS:

ALL APPROVALS FOR THIS DRAWING, ASSOCIATED PARTS LIST(S), AND REVISION CHANGES, ARE CONTROLLED IN THE PRODUCT

PCIE mini Backplane

Clock generator & Reset fanout

File: clock_gen.kicad_sch

Direct 4-lane PCIe

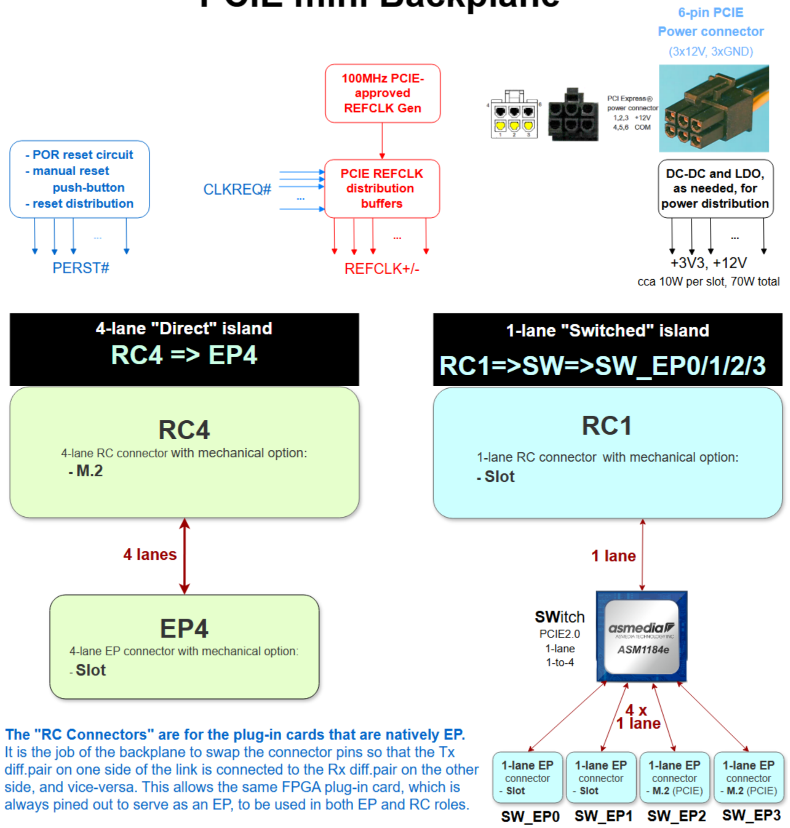
File: direct_PCIe.kicad_sch

PCIe switch

File: pcie_switch.kicad_sch

Switched 4 x 1-lanes PCIe

File: 4_1-lanes_PCIe.kicad_sch



H1 H2 H3 H4 H5 H6

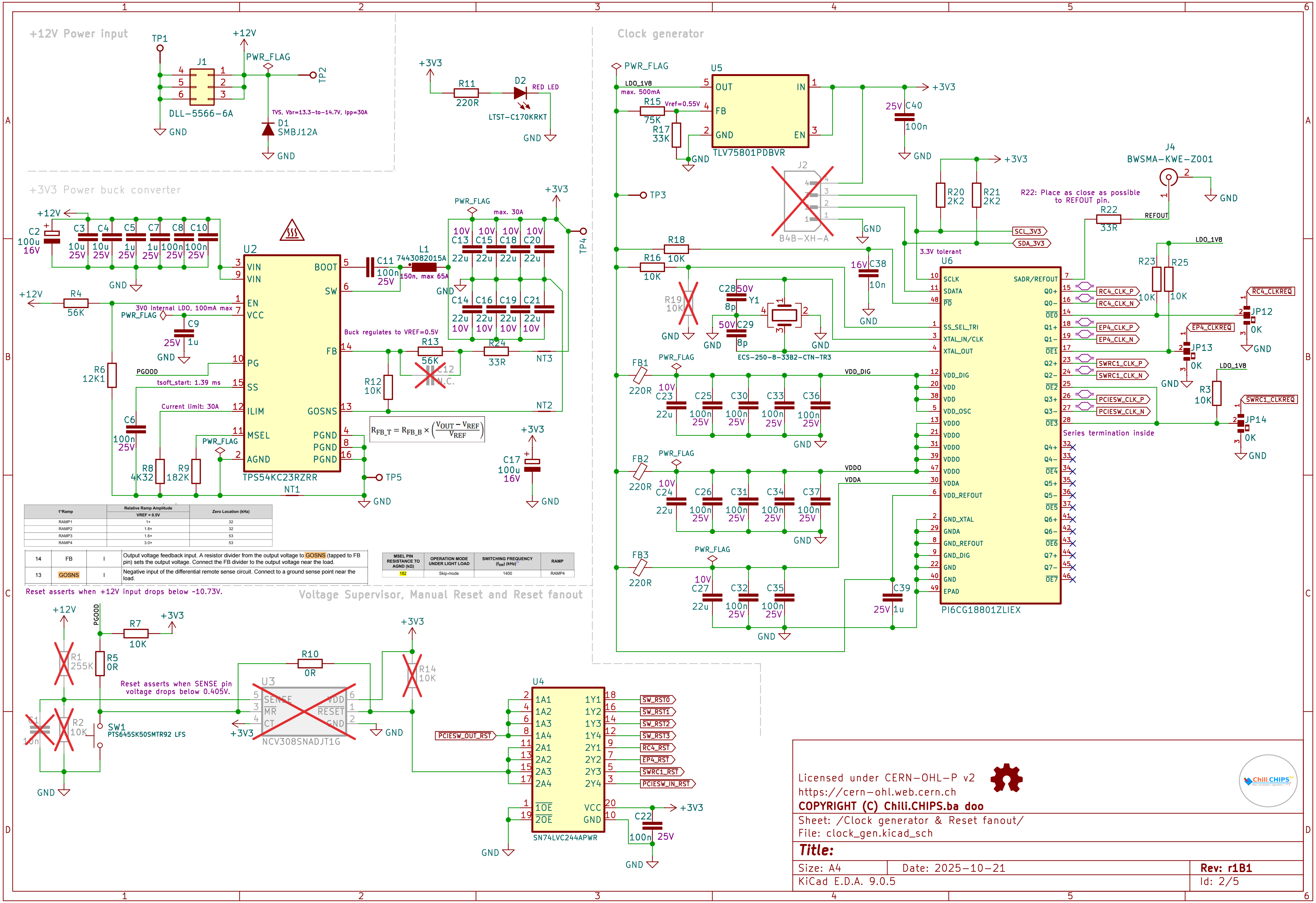
H7 H8 H9 H10 H11 H12

Sheet: /

File: openpci2-backplane.kicad_sch

Title:

Size: A4	Date: 2025-10-21	Rev: r1B1
KiCad E.D.A. 9.0.5		Id: 1/5



1*Ramp	Relative Ramp Amplitude VREF = 0.5V	Zero Location (kHz)
RAMP1	1×	32
RAMP2	1.8×	32
RAMP3	1.6×	53
RAMP4	3.0×	53

14	FB	I	Output voltage feedback input. A resistor divider from the output voltage to GOSNS (tapped to FB pin) sets the output voltage. Connect the FB divider to the output voltage near the load.
13	GOSNS	I	Negative input of the differential remote sense circuit. Connect to a ground sense point near the load.

MSEL PIN RESISTANCE TO AGND (kΩ)	OPERATION MODE UNDER LIGHT LOAD	SWITCHING FREQUENCY (f_{SW}) (kHz)(1)	RAMP
182	Skip-mode	1400	RAMP4

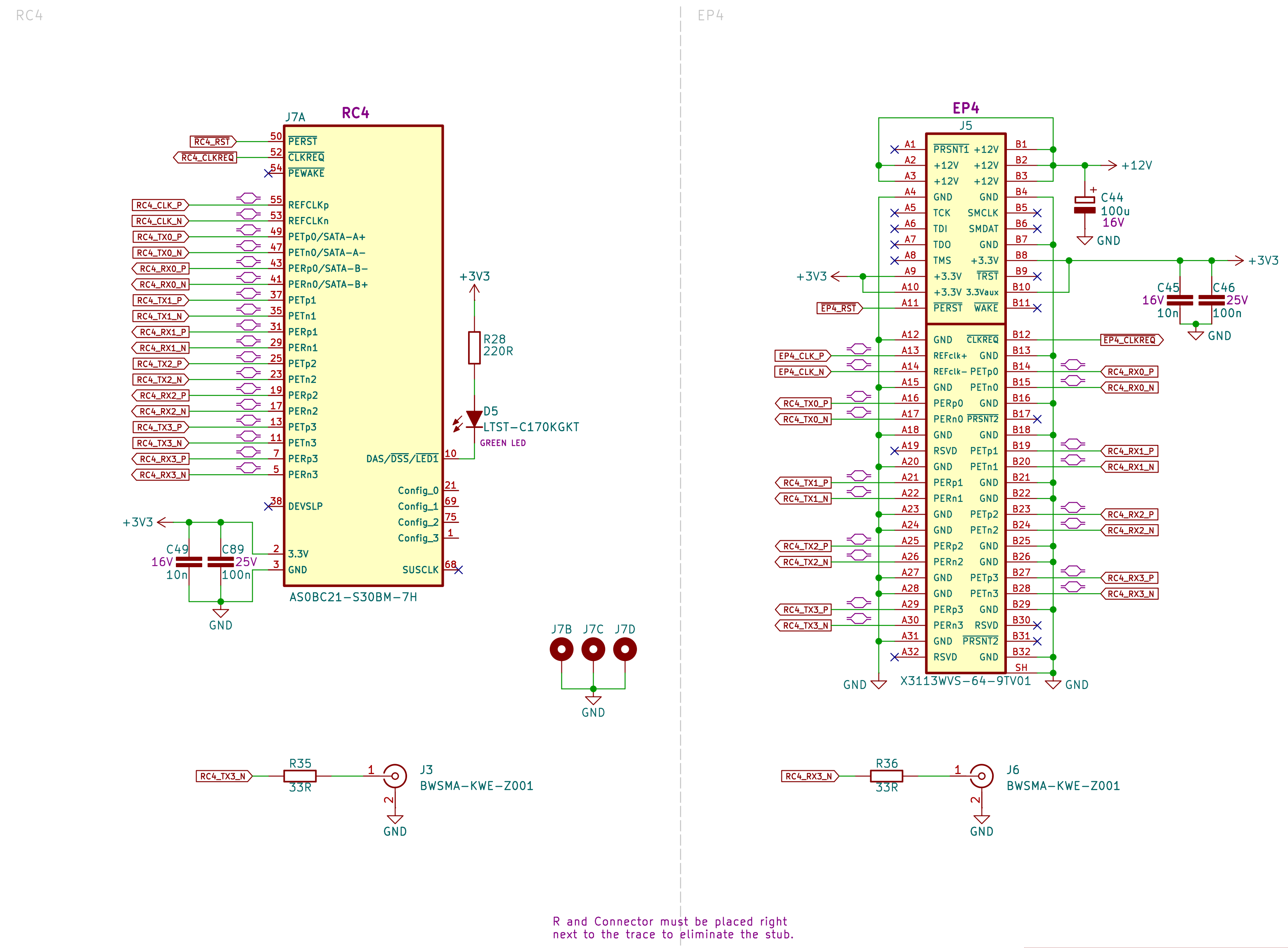
Sheet: /Clock generator & Reset fanout/
File: clock_gen.kicad_sch

Title:

Size: A4 | Date: 2025-10-21 | Rev: r1B1
KiCad E.D.A. 9.0.5 | Id: 2/5

RC4

EP4



PCIEx1 pinout

Pin	Side B	Side A	Description
1	+12 V	~~PRSNT1#~~	Must connect to farthest PRSNT2# pin
2	+12 V	+12 V	Main power pins
3	+12 V	+12 V	
4	Ground	Ground	
5	~~SMCLK~~	~~TCK~~	SMBus and JTAG port pins
6	~~SMDAT~~	~~TDI~~	
7	Ground	~~TDO~~	
8	+3.3 V	~~TMS~~	
9	~~TRST#~~	+3.3 V	
10	+3.3 V aux	+3.3 V	Aux power & Standby power
11	~~WAKE#~~	PERST#	Link reactivation; fundamental reset [25]
Key notch			
12	CLKREQ#[26]	Ground	Clock Request Signal
13	Ground	REFCLK+	Reference clock differential pair
14	HSOp(0)	REFCLK−	
15	HSOn(0)	Ground	Lane 0 transmit data, + and −
16	Ground	HSIp(0)	Lane 0 receive data, + and −
17	~~PRSNT2#~~	HSIn(0)	
18	Ground	Ground	

PCI Express x1 cards end at pin 18

Legend	
Ground pin	Zero volt reference
Power pin	Supplies power to the PCIe card
Card-to-host pin	Signal from the card to the motherboard
Host-to-card pin	Signal from the motherboard to the card
Open drain	May be pulled low or sensed by multiple cards
Sense pin	Tied together on card
Reserved	Not presently used, do not connect

Sheet: /Direct 4-lane PCIe/
File: direct_PCIe.kicad_sch

Title:

Size: A4 | Date: 2025-10-21 | Rev: r1B1
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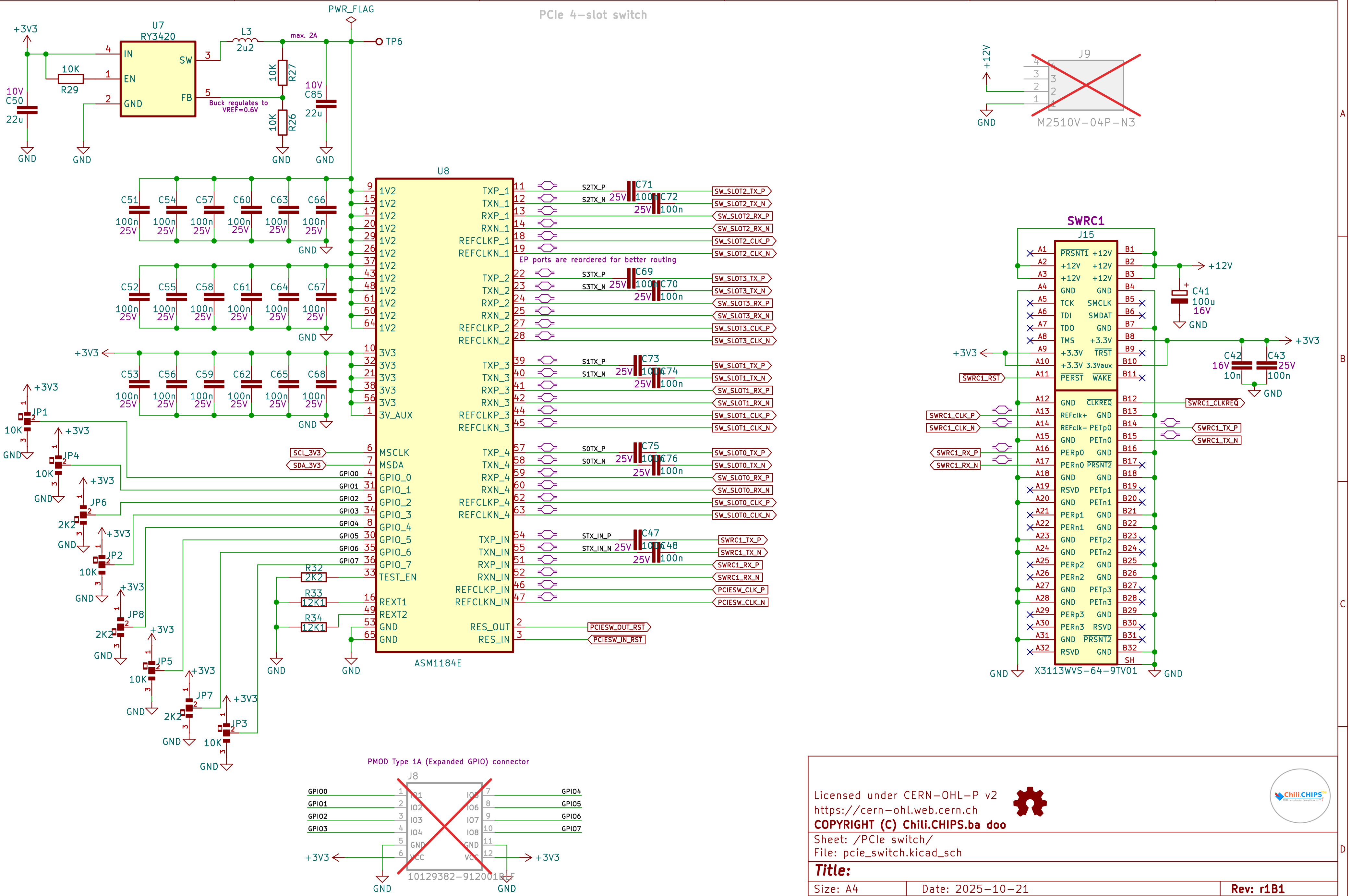
PCIe 4-slot switch

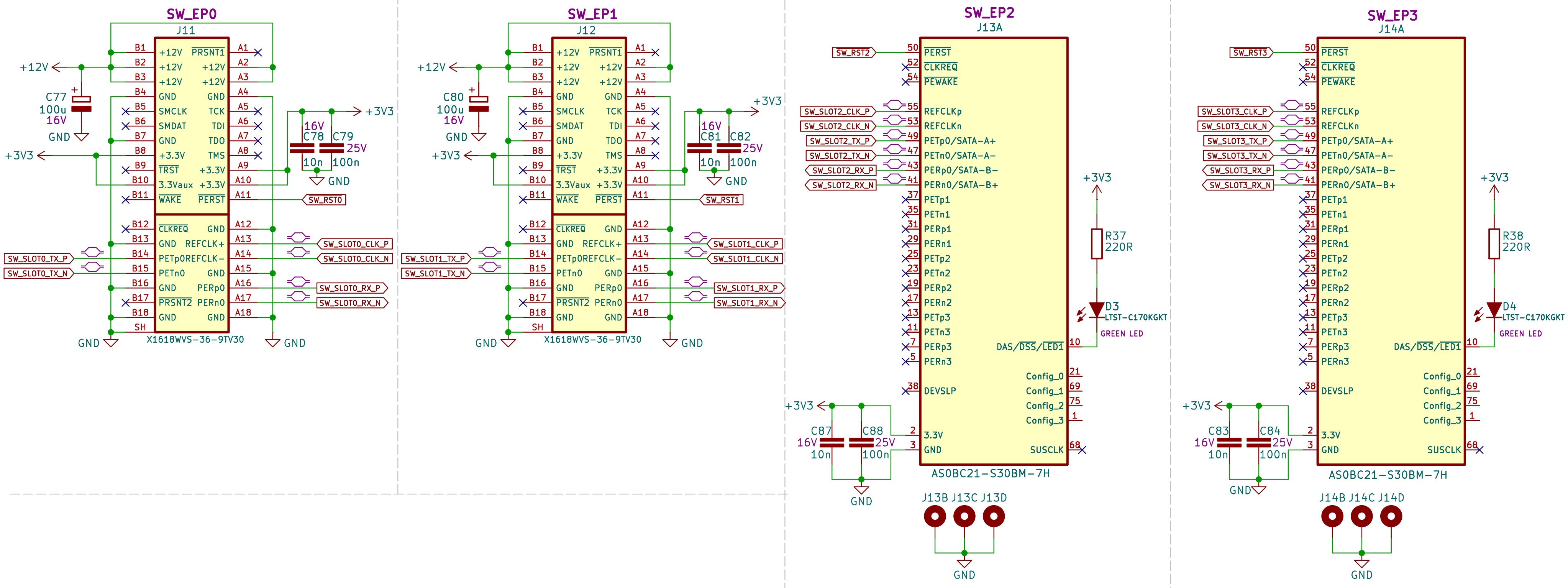


Sheet: /PCIe switch/

File: pcie_switch.kicad_sch

Title:

Size: A4	Date: 2025-10-21	Rev: r1B1
KiCad E.D.A. 9.0.5		Id: 4/5



Sheet: /Switched 4 x 1-lanes PCIe/
File: 4_1-lanes_PCIe.kicad_sch

Title:

Size: A4	Date: 2025-10-21	Rev: r1B1
KiCad E.D.A. 9.0.5		Id: 5/5